

Focus: Master problem visualization, code dry-running, and thinking out loud during interviews.

Approach: Random topic cycling (Mission 250 DSA)—switching topics dynamically rather than staying in a single category.

Deadline:  15th August 2026 (Very difficult target) - 9 questions per day - Decide when to do easy medium hard (3 5 1) or (4 4 1) or (2 5 2) morning/evening/night session

Be able to build a solution in mind as soon as i catch a question name, like many people who are regular at dsa know the problem by the problem's name in their mind i need to get to that At least have a good revision on the question i have solved and come up with their solution as soon as i see them, not wasting time how to revise what i already know and build up on that Better my time and space complexity

For the questions i have already solved but can't come up with the direct solution, anxiety hits me, it's like i am not able to recall what i did earlier to come up with the solution

NOTE: During solving of these questions i did topic wise, so at the point of solving my intuition for that particular topic was way higher as it is now. I still can't segregate which method to best use for a given question

Learning Progress Via Striver's DSA sheet (213/474 solved)

Sorting I:

Concepts Known: Selection, Insertion, Bubble Sort

Intuition: Can get time and space complexities

Visualization: Can visualize

Sorting II:

Concepts Known: Merge, Quick

Intuition: Quick sort is a bit difficult to code and come up with

Visualization: Can visualize

Arrays(14/40 solved)

Concepts Known: Check if the Array is Sorted II, Remove duplicates from Sorted array, Left Rotate Array by One, Left Rotate Array by K Places, Move Zeros to End, Maximum Consecutive Ones, "Find the number that appears once, and other numbers twice.", Sort an array of 0's 1's and 2's, Kadane's Algorithm, Stock Buy and Sell, Rearrange array elements by sign, Set Matrix Zeroes, Rotate matrix by 90 degrees, Print the matrix in spiral manner

Intuition: Easy to medium questions, i can solve the question mostly but not very fluent, like the solution won't strike me as soon as i see the problem, i have solved the above questions but have forgotten what i did there so it's as good as a scratch new problem

Visualization: Easy to medium, strong visualization missing

Binary search(13/32)

Concepts Known: Search X in sorted array, Search insert position, First and last occurrence, Search in rotated sorted array-I, Search in rotated sorted array-II, Find minimum in Rotated Sorted Array, Find peak element, Koko eating bananas, Minimum days to make M bouquets, Find the smallest divisor, Capacity to Ship Packages Within D Days, Kth Missing Positive Number, Search in a 2D matrix

Intuition: Can start and build to the solution but then again don't have much revision or grip on the problems already solved, BS on 1d array is good, on answers is also fine but 2d array is missing

Visualization: Have some sort of visualization but still need to be better at dry running the code

Strings(15/15)

Concepts Known: Remove Outermost Parentheses, "Reverse words in a given string / Palindrome Check", Largest Odd Number in a String, Longest Common Prefix, Isomorphic String, Rotate String, Check if two strings are anagram of each other, Sort Characters by Frequency, Maximum Nesting Depth of the Parentheses, Roman to Integer, String to Integer (atoi), Count Number of Substrings, Longest Palindromic Substring, Sum of Beauty of All Substrings, Reverse every word in a string

Intuition: Strings are not liked by me, sometimes can't get a feel of the question and which hash or something else to use

Visualization: Somewhat but even the solved questions strike to me as new

Linked List(20/31-most is theory)

Concepts Known: Introduction to Singly LinkedList, Insertion at the head of Linked List, Deletion of the head of LL, Find the length of the Linked List, Search in Linked List, Introduction to Doubly LL, Insert node before head in Doubly Linked List, Delete head of Doubly Linked List, Reverse a Doubly Linked List, "Middle of a LinkedList [TortoiseHare Method]", "Reverse a LinkedList [Iterative]", Detect a loop in LL, Find the starting point in LL, Check if LL is palindrome or not, Segregate odd and even nodes in Linked List, Remove Nth node from the back of the LL, Delete the middle node in LL, Reverse LL in group of given size K, Rotate a LL

Intuition: Have the intuition but lack handling of the edge cases, also some clear tricks and best practices are not known

Visualization: I have visualization to some extent but can't visualize the edge cases where the pointers should be

Recursion(12/25)

Concepts Known: Recursive Implementation of atoi(), "Pow(x, n)", Count Good Numbers, Generate Parentheses, Combination Sum, Combination Sum II, Subsets II, Combination Sum III, Letter Combinations of a Phone Number, Word Search, N Queen, Expression Add Operators

Intuition: Can solve on paper but don't know in depth how variables flow through the depths of recursion, which is the base case and edge cases

Visualization: Can't visualize time and space complexities quickly, need some time to see and confirm what variable we are passing and how it influences the layers above or below, even if i come up with a solution i can't visualize it, i can just state it

Bit manipulation(13/18)

Concepts Known: Introduction to Bits and Tricks, Check if the i-th bit is Set or Not, Check if a Number is Odd or Not, Check if a Number is Power of 2 or Not, Count the Number of Set Bits, Set/Unset the rightmost unset bit, Swap Two Numbers, Divide two numbers without multiplication and division, Minimum Bit Flips to Convert Number, Single Number - I, Power Set Bit Manipulation, Count primes in range L to R, "Pow(x,n)"

Intuition: Can't visualize and don't have a grip on the tricks since i don't come across them regularly, also lack much of the theoretical knowledge

Stacks and Queues(20/30):

Concepts Known: Implement Stack using Arrays, Implement Queue using Arrays, Implement Stack using Queue, Implement Queue using Stacks, Implement stack using Linked List, Implement queue using Linked List, Check for balanced paranthesis, Implement Min Stack, Infix to Postfix Conversion using Stack, Prefix to Infix Conversion, Prefix to Postfix Conversion, Postfix to Prefix Conversion, Postfix to Infix Conversion, Infix to Prefix Conversion, Next Greater Element, Next Greater Element 2, Nearest Smaller Element, Sum of subarray minimums, Asteroid Collision

Intuition: Don't have a strong understanding of these, when to use monotonic stack which is very very important

Visualization: can't tell the time and space complexities on the go, visualization is also hard for me to how to come up with the solution

Sliding Window And 2 Pointers(6/12)

Concepts Known: Longest Substring Without Repeating Characters, Max Consecutive Ones III, Longest Repeating Character Replacement, Binary Subarrays With Sum, Count number of Nice subarrays, Number of Substrings Containing All Three Characters

Intuition: Very weak on when to increment and decrement left and right pointers

Visualization: Can't dry run easily and need to be fluent with coming up with the time and space complexities

Heaps(7/17):

Concepts Known: Heaps (Theory Video), Implement Min Heap, Check if an array represents a min heap, Convert Min Heap to Max Heap, Merge K sorted Lists, Task Scheduler, Hand of Straights

Intuition: Don't know how and when to connect this into my problem, i think it can be solved without heaps as well so don't have an intuition on when to use this

Visualization: Could be better specially in case of complexity derivations

Greedy algorithms(6/15):

Concepts Known: Assign Cookies, Lemonade Change, Valid Paranthesis Checker, Jump Game - I, Jump Game II, Insert Interval

Intuition: I do have intuition but not the best ones usually, and have a difficulty in coding my intuition

Visualization: Medium level visualization i would say but could be a lot better in the future

Binary trees and BST (0/54):

I know few concepts but haven't coded them yet

Graphs(10/53-mostly theory):

Concepts Known: Introduction to Graph, Graph Representation | C++, Graph Representation | Java, Connected Components, Traversal Techniques, DFS, Number of provinces, Rotten Oranges, Flood fill algorithm, Detect a cycle in an undirected graph

Intuition: Can identify which of the traversals to best use mostly, but still have a lack of concepts methods and fairly new concept to me just know how to write basics bfs dfs not more, haven't able to deeply understand the process of coming up with a solution and lose track mostly because i am not aware of what other things we could do

Visualization: Space and time complexities are very difficult, can visualize solution but can't initially think of what to do, even if i do, i can't code that, i have the idea but not the implementation mostly

Dynamic Programming(17-55)

Concepts Known: Introduction to DP, Climbing stairs, Maximum sum of non adjacent elements, House robber, Grid Unique Paths : DP on Grids (DP8), Unique paths II, Minimum Falling Path Sum, Partition equal subset sum, Assign Cookies, "Minimum Coins (DP - 20)", Target sum, "Coin Change 2 (DP - 22)", Longest palindromic subsequence, Minimum insertions or deletions to convert string A to B, Edit distance, Best time to buy and sell stock, Best time to buy and sell stock II

Intuition: No not at all, can't come up with the recurrence relation, intuition is very hard to come up with, but once i have it i can code it all right, but really don't understand half of the intuitive approach and how to actually come up with own, like i can come up with a new intuition when i am in a flow and have already solved plenty of DP just now

Visualization: Very difficult to dry run and get the time and space complexities